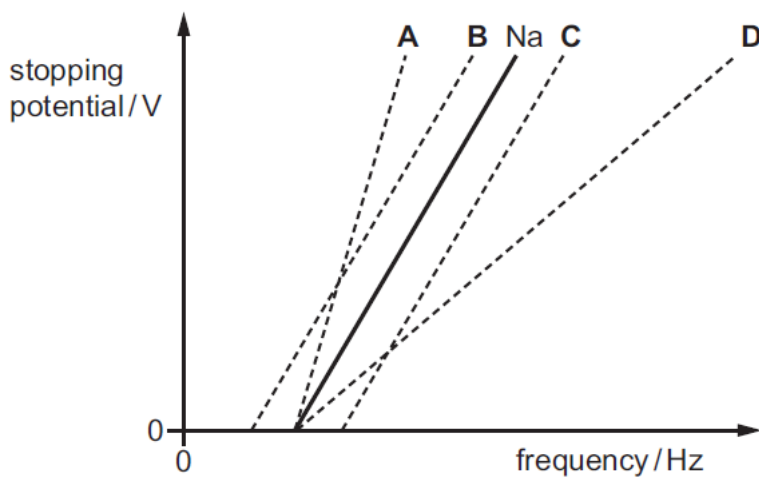


- 27 In a photoelectric cell experiment, sodium (Na) metal is exposed to electromagnetic radiation of different frequencies and the potential required to stop the photoelectrons is measured.

The graph uses a solid line, marked Na, to show the results of the experiment.

Which line represents the results for the photoelectric cell using a metal that has a larger work function than sodium?



Ans: C

$$hf = \Phi + eV$$

$$V = \left(\frac{h}{e}\right)f - \frac{\Phi}{e}$$

$$\Phi \uparrow, -\frac{\Phi}{e} \uparrow, \text{ negative y-intercept } \uparrow$$

28 The diagram shows the lowest four energy levels in the hydrogen atom. The energy levels of